Exact-Wick (anchored) two-shell no-condensate at the B1 point: the off-diagonal bracket does not overturn the diagonal result

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

twoshell-continuum-bound v1.0 established the two-shell no-condensate at the B1 point $r_{\text{bare}} = 0.219$ for the DIAGONAL free energy and named the exact-Wick off-diagonal bracket as the residual: the anchored free energy is $\Delta F_{\text{anchored}} = \Delta F_{\text{diag}} + \text{bracket}$ with bracket $= F_{\text{exact}} - F_{\text{diag-basis}}$, and the bracket is negative, so diagonal positivity does not by itself imply anchored positivity. This note evaluates the bracket with the EXACT slogdet engine and settles the anchored question.

2. Engine reuse and validation

The Math432 exact-Wick engine (slogdet of the dressed Bloch Hessian over a BCC k-mesh) is REUSED without re-transcription: the legacy script is read, truncated before its module-level scan, its $m424$ import path is repointed, and the result is imported as a module. The engine reproduces Math432's RECORDED brackets at $r = 0.005$ to 5×10^{-8} (two anchors), so the reuse is faithful before any $r = 0.219$ evaluation.

3. Exact-Wick no-condensate at the B1 point ($r = 0.219$)

Over the Math432 scan domain ($A_1 \in [0, 0.16]$, $A_2 \in [-0.16, 0.16]$, $M/M_R \in \{0.7, 1.0, 1.3\}$, $11 \times 11 \times 3$):

- **no exact condensate**: $\min_{(A_1,A_2)\neq(0,0)} \Delta F_{\text{anchored}} = +6.7 \times 10^{-4} > 0$ (at small A_1). The off-diagonal bracket does NOT open a sub-Reading-H valley.
- **near origin (continuum)**: the bracket is $O(A^4)$ – the off-diagonal $W \sim$ amplitude^2, so bracket $\sim \mathrm{Tr}[\cdot] \sim W^2 \sim A^4$; numerically $|\text{bracket}|(A{=}0.005) = 3.9 \times 10^{-8}$. Hence the anchored $(0,0)$ Hessian EQUALS the diagonal one, $\kappa_{\{110\}} = 5.16$, $\kappa_{\{200\}} = 3.86$ (both > 0): anchored positive-definiteness near the origin is a CONTINUUM statement.
- **large amplitude**: $|\text{bracket}|$ grows to 4.4×10^{-2}, but there $\Delta F_{\text{diag}} \sim 10^{-1}$, so $\Delta F_{\text{anchored}}$ stays positive; $|\text{bracket}|$ and ΔF_{diag} do not subtract uniformly (both peak at large A, while the thinnest anchored value is at small A where the bracket is negligible).

4. What this closes and what remains

CLOSED: the exact-Wick (anchored) two-shell no-condensate at the B1 point, at GRID grade in the bulk and CONTINUUM grade near the origin (the $O(A^4)$ bracket leaves the PD Hessian intact). This settles the physical question the bracket residual posed: the off-diagonal correlations do not overturn the selection. REMAINING (a refinement): a curvature-chord continuum bound on the exact anchored BULK surface (a finer exact scan), to upgrade the bulk from grid to node-free. The gate stays OPEN pending that refinement and operator sign-off; the substantive obstruction is removed.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) *"The bulk is still a grid statement."* VALID and named: the $11 \times 11 \times 3$ exact scan certifies the nodes; a node-free bulk bound needs the curvature-chord on the anchored surface (heavier exact scan). The near-origin region – where the diagonal margin is thinnest – is already continuum (PD Hessian, $O(A^4)$ bracket), so the grid-vs-continuum gap is in the bulk where the margin is large ($\Delta F_{\text{anchored}} \gtrsim 6.7 \times 10^{-4}$).

(β) *"Is the neutered engine the same as Math432?"* DISMISSED: it reproduces Math432's recorded brackets to 5×10^{-8}; the import is the SAME source truncated before the scan, not a re-transcription.

(γ) *"Could a finer M-grid find a negative anchored value?"* DISMISSED at strong-evidence grade: the anchored min is at small A where the bracket is $O(A^4) \sim 10^{-8}$, so $\Delta F_{\text{anchored}} \approx \Delta F_{\text{diag}} > 0$ there irrespective of M; the bracket cannot turn the small-A region negative.

Quantitative sanity check (mandatory): magnitude – $|\text{bracket}|(h) = 3.9 \times 10^{-8}$ vs the curvature scale $\kappa h^2/2 \sim 6 \times 10^{-5}$ ($O(A^4)$ confirmed); sign-direction – the bracket is negative but the anchored min $+6.7 \times 10^{-4}$ is positive; limit-case – the engine reproduces Math432 at $r = 0.005$ to 5×10^{-8}; consistency – the anchored $(0,0)$ eigenvalues 5.16/3.86 match the diagonal ones of twoshell-continuum-bound v1.0 (the bracket does not shift the Hessian).

6. Result footer

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Result ID:              B1-RH-ENUM / exact-Wick anchored two-shell no-condensate
                        at the B1 point (closes the bracket residual)
Precise statement:      At r_bare=0.219, the anchored two-shell {110}+{200} free
                        energy dF_anchored = dF_diag + (F_exact - F_diag_basis)
                        has min over (A1,A2)!=(0,0) = +6.7e-4 > 0 over the Math432
                        scan domain (11x11x3); the bracket is O(A^4) near origin
                        (|bracket|(0.005)=3.9e-8), so the anchored (0,0) Hessian
                        = diagonal (kappa_{110}=5.16, kappa_{200}=3.86, PD).
                        Engine reused (Math432 neuter-import) validated to 5e-8.
Scope:                  Two-shell {110}+{200}, EXACT-Wick anchored free energy,
                        B1 point. Bulk = grid; near-origin = continuum (PD).
Dependencies:           Math432 engine (neuter-imported); Math424_AddA (gap/loop);
                        twoshell-continuum-bound v1.0 (diagonal bound it closes).
Evidence grade:         EXECUTED (twoshell_anchored_bracket.py 7/7); exact-Wick
                        grid + near-origin continuum.
Reproduction command:   python codes/vacuum/twoshell_anchored_bracket.py
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Expected output:         7/7 PASS; min anchored +6.7e-4; |bracket|(h)=3.9e-8;
                         anchored kappa_{110}=5.16, kappa_{200}=3.86.
Falsification gate:      An anchored surface point with dF<0 at r=0.219; a bracket
                         that is NOT O(A^4) near origin; an anchored (0,0) Hessian
                         that differs from the diagonal one.
Tier before / after:     No tier/gate flip. ESTIMATOR-UPGRADE: the exact-Wick
                         two-shell bracket residual is CLOSED (grid + near-origin
                         continuum); gate stays OPEN pending the bulk anchored
                         continuum refinement + operator sign-off.
No-overclaim statement:  Bulk is a GRID statement on the exact anchored surface;
                         near-origin is continuum (PD, O(A^4) bracket). NOT a
                         node-free bulk theorem. The substantive bracket
                         obstruction is removed; a finer exact scan upgrades the
                         bulk to continuum.
Next required action:    Curvature-chord continuum bound on the exact anchored bulk
                         surface (finer exact scan); then ESTIMATOR-UPGRADE closure
                         is an operator decision.
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